# HOW TO LEARN GEAR´S @MPS

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## Values for Experiment

- Tra_numGear → Current gear information
- Tra_uRawGear → Gear sensor Raw voltage
- Tra_uGearPosnAry → Array for sensor voltages of specific gear position
- Tra_flgGearPosnTblFull → Flg for stage one is finished
- Tra_flgGearPosnRdyChk → Flg for stage two is finished
- Tra_flgSetCorrBitGearPosn → Bit string for found and checked gears during stage three
- Tra_flgRdyChkTbl → Flg for average value correct, for checking during stage three
- Tra_flgEvlnFinshd → Flg for evaluation of all the gears is finished
- Tra_flgEvalErrSet → Flg for acquired voltage is not found in table during stage three
- Tra_flgGearSnsrTblInvld → Flg for gear learning not done or threshold table is damaged

- Clth_st → Clutch information (Trigger in stage on and three)
- I14230_flgGearValRst_C → CW for controlling tester request for reset of gear values



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# How to learn gear´s @MPS
## Step I

If the gear learning is not done (**Tra_flgGearSnsrTblInvld = true**) the gear learning procedure is started automatically. While learning the engine doesn't need to run.

i. Shift to every gear (6 gear + neutral) and trigger the signal acquisition in each gear by pulling the clutch (**Clth_st = true**). The function collects 10 average values over 100 static samples over one second. Be sure that the gear leaver isn´t touched after triggering. Cross check if the new voltage in the array **Tra_uGearPosnAry** is matching with **Tra_uRawGear** after triggering.

ii. If the difference of the min/max value of the collected 10 values of each gear is not above **Tra_uGearVoltMinMaxDiff_C** and the value is not already collected, the average of the 10 values is written in the next free place of the table **Tra_uGearPosnAry**. If a similar value already exists in the table a gear change is required for learning the value for the next gear

iii. If each gear has a value in the table (**Tra_flgGearPosnTblFull = true**), stage two is started**.**



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## Step II

In stage two the values in the table **Tra_uGearPosnAry** are arranged in ascending order. If this is successfully done **Tra_flgGearPosnRdyChk** will become true.



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## Step III

In stage three each gear has to be selected at least once and checked in the display for correctness.

i. After a gear is selected and triggered with **Clth_st = true** the ECU checks if the value is already in the table **Tra_uGearPosnAry**. If found, the ECU waits for the time stored in **Tra_tiSetDlyTilNxtGear_C** and set corresponding bit of **Tra_flgSetCorrBitGearPosn**.

ii. Perform the verification for each gear (6 gear + neutral) and check if the corresponding bit is set in **Tra_flgSetCorrBitGearPosn**. As soon all gears are checked (**Tra_flgSetCorrBitGearPosn = 127**) the flag **Tra_flgEvlnFinshd** will become true. If any gear isn´t found and couldn't be verified (**Tra_flgEvalErrSet = true**), the whole gear learning procedure is restarted.

iii. If all seven gears are found/verified, the array will be stored in the EEPROM. For that, a fully finished afterrun is required. After successful learning and afterrun the learned values should remain stored in **Tra_uGearPosnAry**.



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